Robed Beauvile-Owen, Ph.D.

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EXPERIENCE

Founder & Data Scientist — *AniPoint Site Research* Pittsburgh, PA | May 2026 - Present

- Founded a location intelligence consultancy for retail and service businesses, identifying a $1.5M expansion opportunity.
- Built a revenue driver tree (traffic × capture rate × ticket price) and a margin model that turned client profit goals into break-even targets.
- Estimated the incremental revenue of new sites with Double ML and causal forests on a 12-year county panel, checked with difference-in-differences.
- Back-tested demand predictions against real sites, then built low/base/high scenarios and Monte Carlo risk models (probability of loss, VaR) to show uncertainty.
- Built tested Python pipelines (150+ unit tests) that cut analysis time by 91%, and documented each method for reuse.

Senior Data Scientist — *Limetree* Dallas, TX (Remote) | September 2024 - April 2026

- Owned the acquisition KPI framework (gross lead rate, member rate, cost per member) with strategy, finance, and engineering leadership.
- Redefined gross lead rate (GLR) for a new client with missing lead data, working with data engineering to set lead date, last-response credit, and broker flags.
- Built a causal graph (DAG) showing GLR drives member rate, which drives cost per member, focusing modeling and tests on GLR.
- Built uplift and segmentation models used in 87% of campaigns, cutting ad spend by $10K per campaign, and found the segment driving one client's GLR.
- Built an in-house experimentation tool used to validate lift and set campaign budgets, with CUPED cutting metric variance by 20%.
- Linked the monthly retention survey to member behavior, finding that claims and ticket complaints drove churn.
- Owned monthly monitoring of up to 10 production models for leadership and a company-wide self-service dashboard built on Snowflake and Databricks.

Founding Machine Learning Engineer — *eParts Services LLC* Pittsburgh, PA | September 2023 - September 2024

- As the first ML hire, defined success metrics and drift monitoring for search and pricing models, and mentored the team.
- Shipped a regression pricing API (C#, Azure, 10,000+ daily requests) and cut search dissatisfaction by 12%.

Data Scientist — *Limetree* Dallas, TX (Remote) | April 2022 - September 2023

- Defined and reported the core campaign metrics (GLR, member rate, cost per member).
- Built XGBoost response models that lifted response rates by 3%, explained to stakeholders with SHAP.
- Automated Power BI reporting, saving 60 minutes daily, and improved geographic targeting accuracy by 20%.

Graduate Researcher & Teaching Assistant — *University of Pittsburgh* Pittsburgh, PA | August 2017 - May 2022

- Produced 3 original results in optimization and operator theory, and taught statistics to classes of 75-200 students.

TECHNICAL PROJECTS

Customer LTV & Churn Survival Modeling github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Showed when LTV = ARPU / churn breaks down, using 1M simulated customers and survival curves.

Mobile Game Retention A/B Test github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Tested a change on 1- and 7-day player retention with sample sizing and significance testing.

Medicare Claims Intelligence & Fraud Detection github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Built a dbt warehouse on 790K claims, a fraud detector with 81% precision, and forecasts 41% more accurate than a seasonal benchmark.

Starbucks Rewards: Predicting Offer Responses github.com/beauvilerobed/predict-consumer-responses

- Predicted how rewards members respond to offers with XGBoost on AWS SageMaker.

EDUCATION

Ph.D. in Mathematics — *University of Pittsburgh* 2017 - Dec 2023

Dissertation: A Minimal Invariant Set, Frames, and Operators

B.S. in Mathematics — *Florida State University* 2014 - 2017

Minor: Philosophy | **GPA:** 3.90

TECHNICAL SKILLS

Metrics: KPI Trees, Metric Definitions, Funnel & CAC, Retention, LTV / ARPU, Scenario Planning, Forecast Backtesting

Causal & Experimentation: A/B Testing, CUPED, Double ML, Causal Forests, Difference-in-Differences, Uplift Modeling, DAGs

Modeling: Regression, XGBoost, Time Series Forecasting, Survival Analysis, Monte Carlo Simulation, SHAP

Tools: SQL, Python, Snowflake, Databricks, Spark, dbt, AWS, Azure, Power BI, Tableau, Git